

JavaScript Functions

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Functions are Code Blocks

A JavaScript function is a block of code designed to perform a particular task.

A JavaScript function is executed when "something" invokes it (calls it).

```
<!DOCTYPE html>
<html>
<body>

<h2>JavaScript Functions</h2>
<p>From w3schools.com, Experiment by Teeratus_R</p>

<h4>Functions are Code Blocks</h4>
<p id="demo"></p>

<script>
function myFunction() {
  document.getElementById("demo").innerHTML = "Hello World!";
}
myFunction();
</script>

</body>
</html>
```

Example 1

Result [View Example](#)

Functions Run When You Call Them

Functions run when you call them. You can call the function by its name:

```
<!DOCTYPE html>
<html>
<body>

<h2>JavaScript Functions</h2>
<p>From w3schools.com, Experiment by Teeratus_R</p>

<h4>Functions Run When You Call Them</h4>
<p id="demo"></p>

<script>
function myFunction(a, b) {
    return a * b;
}
let result = myFunction(4, 3);
document.getElementById("demo").innerHTML = "4 * 3 = " + result;
</script>

</body>
</html>
```

Example 2

Result [View Example](#)

JavaScript Function Syntax

A JavaScript function is defined with the `function` keyword, followed by a **name**, followed by parentheses `()`.

Function names can contain letters, digits, underscores, and dollar signs (same rules as variables).

The parentheses may include parameter names separated by commas:

```
(parameter1, parameter2, ...)
```

The code to be executed, by the function, is placed inside curly brackets: `{}`

```
function name(parameter1, parameter2, parameter3) {
    // code to be executed
}
```

```
<!DOCTYPE html>
<html>
<body>

<h2>JavaScript Functions</h2>
<p>From w3schools.com, Experiment by Teeratus_R</p>

<h4>JavaScript Function Syntax</h4>
<p id="demo"></p>

<script>
// Function is called, the return value will end up in x
let x = myFunction(4, 3);
document.getElementById("demo").innerHTML = x;

function myFunction(a, b) {
  // Function returns the product of a and b
  return a * b;
}
</script>

</body>
</html>
```

Example 3

Result [View Example](#)

A Function Can Be Used Many Times

Functions can be reused to perform the same task with different values:

```
<!DOCTYPE html>
<html>
<body>

<h2>JavaScript Functions</h2>
<p>From w3schools.com, Experiment by Teeratus_R</p>

<h4>A Function Can Be Used Many Times</h4>
<p id="demo1"></p>
<p id="demo2"></p>
<p id="demo3"></p>

<script>
function toCelsius(fahrenheit) {
    return (5/9) * (fahrenheit - 32);
}

document.getElementById("demo1").innerHTML = "32°F = " + toCelsius(32) + "°C";
document.getElementById("demo2").innerHTML = "68°F = " + toCelsius(68) + "°C";
document.getElementById("demo3").innerHTML = "100°F = " + toCelsius(100) + "°C";
</script>

</body>
</html>
```

Example 4

Result [View Example](#)

Local Variables

Variables declared within a JavaScript function become **LOCAL** to the function.

Local variables can only be accessed from within the function:

```
<!DOCTYPE html>
<html>
<body>

<h2>JavaScript Functions</h2>
<p>From w3schools.com, Experiment by Teeratus_R</p>

<h4>Local Variables</h4>
<p id="demo"></p>

<script>
function myFunction() {
  let carName = "Volvo";
  document.getElementById("demo").innerHTML = "Inside function: " + carName;
}
myFunction();
// carName is NOT accessible outside the function
</script>

</body>
</html>
```

Example 5

Result [View Example](#)

Functions Used as Variables

Functions can be used the same way as variables, in all types of formulas, assignments, and calculations:

```
<!DOCTYPE html>
<html>
<body>

<h2>JavaScript Functions</h2>
<p>From w3schools.com, Experiment by Teeratus_R</p>

<h4>Functions Used as Variables</h4>
<p id="demo"></p>

<script>
function myFunction(a, b) {
  return a * b;
}
// Function result used directly as a variable
document.getElementById("demo").innerHTML = "5 * 3 = " + myFunction(5, 3);
</script>

</body>
</html>
```

Example 6

Result [View Example](#)

Document

Document in project

You can [Download PDF](#) file.

Reference

- [W3Schools JavaScript Functions](#)